Uncertainty-Aware Graph Self-Supervised Learning for Hyperspectral Image Change Detection

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***Abstract*— Deep learning-based hyperspectral image (HSI) change detection (CD) has been a research hotspot. However, the high dimensionality and the limited training samples make HSI-CD difficult to implement. Besides, the data uncertainty inherited from HSIs is often neglected. To deal with these issues, this article presents an uncertainty-aware graph self-supervised learning (UA-GSSL) approach for unsupervised HSI-CD, which allows encoding not only the spectral and topological structure attributes but also the data uncertainty into learned feature representation for downstream CD task. Specifically, spectral and spatial correlations in HSIs are first characterized via graph model. Then, based on the constructed spectral–spatial graph models, novel node- and edge-level data augmentations are devised to enrich the contrastive sample pairs. Third, a graph-based dual-branch self-supervised learning (SSL) contrastive network is designed to maximize the mutual information between a pair of low-dimensional feature embeddings. Fourth, a simple but effective UA loss function is dedicatedly formed to encourage reliable features to play the more dominant role in feature representation. Finally, a change map is produced using the learned features. Experimental results obtained on four real HSI datasets sufficiently demonstrate that the UA-GSSL achieves remarkable results compared to 12 state-of-the-art (SOTA) methods. The source code of this article can be downloaded from https://github.com/vickyiiiii /UA-GSSL.**

***Index Terms*— Change detection (CD), contrastive learning (CL), graph embedding, hyperspectral image (HSI), self-supervised learning (SSL), uncertainty aware (UA).**

I. Introduction

CHANGE detection (CD) as an important application of identifying differences among multitemporal remote sensing images is the most common way to timely monitor large-scale change information about the Earth's surface. Hyperspectral image (HSI) with tens to hundreds of continuous and subdivided spectral bands in the ultraviolet, visible, and near-infrared regions of electromagnetic spectrum can provide richer spectral information than other remote sensing images [1]. Thus, HSIs have been applied to many fields, such as classification [2], CD [3], and so on. HSI-based CD (HSI-CD), which targets at detecting the precise land-cover transformation, can uniquely offer subtle and various changes. In the last 20 years, HSI-CD has been widely used in many important fields, which has shown great potential ranging from environmental protection, land coverage updating to precision agriculture, and disaster assessment [4], [5].

Many state-of-the-art (SOTA) algorithms for HSI-CD have been proposed [4], [5], [6]. The early HSI-CD methods generally obtain the CD map through a similarity measure between bi-temporal HSIs [7], [8]. Thus, how to define a generalized similarity measure is a main problem for these traditional methods. However, due to high-correlation and high-dimension characteristics of hyperspectral data, it is not a piece of cake to extract useful information from a large amount of redundant data. To solve the problems of high-dimensional space and data redundancy, principal component analysis (PCA) was adopted to reduce the dimensionality of HSIs, and then, change vector analysis (CVA) was used to separate the reduced spectral features and produce the change map [9], [10], [11]. Though being fast in speed and working well in reducing the data redundancy of HSIs, these methods often perform unsatisfactorily. The main reason is that the pixels in the HSI are highly correlated, while these HSI-CD algorithms only treat the pixel as the basic unit and ignore the spatial context information among them, which causes these methods inefficient. To handle the complex information of HSIs more thoroughly and effectively, a popular way is to make full use of both the spectral characteristics and spatial context features simultaneously to reduce the uncertainty of high redundancy and relieve the complexity of high correlation. There are many HSI-CD algorithms combining spectral and spatial information to improve CD accuracy [12], [13]. For example, Hou et al. [12] proposed a three-order tucker decomposition approach to make full use of both spectral and spatial information for HSI-CD. Wu et al. [13] proposed a subspace-based HSI-CD by measuring spectral changes. This method regarded the observed pixel in the post-event image as target and constructed the background subspace using the corresponding pixel in the pre-event image. These spatial–spectral joint techniques take advantage of the spatial context information contained in HSIs while using abundant spectral information.

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Traditional methods only build upon primitive shallow features, and the detection performance is poor when facing some complex scenes. To further solve these problems, deep learning because of its powerful ability to represent learning has become an effective solution. The deep learning-based methods that try to effectively extract the abstract features while reducing the dimensions of the HSIs provide a potential way to exploit both the spectral information of pixels and their neighboring correlations in the process of CD. For instance, convolutional neural network (CNN) has been widely used to extract spectral, spatial, and temporal information for HSI-CD [14], [15]. Although CNN has been proven to be effective in HSI-CD, traditional CNN-based methods can only perform convolution to learn spectral–spatial features in the Euclidean space, which is not capable of fully maintaining the topological structural characteristic contained in input HSIs. Recently, graph convolutional network (GCN) has attracted increasing attention in HSI-CD [16], [17] and HSI classification [18], [19]. As the structure of GCN enables it to perform convolution operations on irregular areas and effectively maintain the class boundary, GCN provides an effective solution to flexibly capture both the spectral characteristics and spatial information of HSIs by transforming the feature information from the adjacent nodes in the graph model. Hence, GCN is especially suitable for solving the CD problem. Although GCN has its unique advantages in remote sensing image processing, it still faces some difficulties when applied to HSI-CD. First, the high dimensionality of HSI makes it ineffective to directly use the GCN to construct the graph structure at the pixel level [16]. Second, the graph similarity distance measure is computationally intractable and it is not useful in practical HSI-CD applications. Third, the limited number of available training samples and the laborious high-dimensional data labeling impede the application of GCN under certain conditions. Finally, GCN cannot provide the confidence score for the predicted labels, which is desirable to know how much to trust the model output for downstream task.

For the first and second issues listed above, a common approach is to embed graph on a more conventional geometric space while preserving relevant graph structural properties. Graph embedding [20], as an effective dimensionality reduction method, tries to convert graph data into a low-dimensional space vector while maintaining the graph structural information in original hyperspectral data. In this way, graph-related algorithms can then be computed efficiently. Hence, it allows the HSI processing to benefit from the computational efficiency of the SOTA statistical models and tools along with the convenience and representational power of classical symbolic representations.

For the third and fourth problems mentioned above, learning from unlabeled data based on attributed graph may be a good way out. Self-supervised learning (SSL) [21], which has become a promising and trending learning paradigm for many tasks in both computer vision and natural language processing, achieves impressive improvements on various tasks. SSL aims to extract representative informative knowledge through well-designed pretext tasks to generate pseudo labels without relying on manual labels. Hence, SSL provides novel insights into reducing the dependency on annotated labels and enables the training on massive unlabeled data. In [22] and [23], contrastive learning (CL), as a popular method of SSL, was employed for HSI classification and satisfied performance was achieved. Graph SSL (GSSL), which is a special SSL task, usually involves learning the graph structure or the context information in the data without any annotated labels. GSSL, because of its nature of learning graph embeddings, also offers a possible solution for uncertainty quantification estimation in terms of both attribute uncertainty and structure uncertainty.

Given the consideration of problems and concerns mentioned above, an uncertainty-aware graph SSL (UA-GSSL) framework for unsupervised HSI-CD is proposed in this article. The proposed method UA-GSSL allows encoding not only the spectral and topological structure attributes but also the data uncertainty into learned feature representation for downstream CD task. The UA-GSSL consists of four main parts: graph construction, data augmentation, graph embedding, and graph CL. Specifically, since the structural information of image can be intuitively expressed in the form of graph model, a spectral–spatial graph model is adopted to fully exploit the informative spatial and spectral features of multitemporal HSIs. Then, based on the constructed spectral–spatial graph models, two novel node- and edge-level data augmentations are proposed to generate a set of graph instances and expand the contrastive sample pairs, which is the footstone for SSL. To deal with the high computation burden, a graph-based SSL contrastive network with a dual-branch structure is designed to learn representative and robust low-dimensional feature vectors from high-dimensional data formed in both nodes and edges via graph embedding and CL. Fourth, considering the problem of uncertainty in learned features, for example, when the two nodes are regarded as the nearest neighbors via a similarity measure and are connected by an edge in graph model, while the labels of two nodes are different from each other due to the problem of the same spectral but for different objects, a simple but effective uncertainty-aware (UA) loss function is dedicatedly formed to encourage the discriminative features to play the more dominant role in learned features. Finally, the change map is produced by classifying the learned feature vectors into two classes, the changed and the unchanged.

In particular, the main contributions of our work can be summarized as follows.

1) A novel GSSL-based unsupervised HSI-CD method is proposed. As far as we know, this is the first work that introduces GSSL into the community of remote sensing.
2) Graph-based node- and edge-level data augmentations are devised to enrich the contrastive sample pairs, which is helpful for taking advantage of unlabeled data.
3) Uncertainty awareness is highlighted in feature learning, which encourages the more discriminative and reliable features to play the more dominant role in downstream HSI-CD task.

The remainder of this article is organized as follows. Section II reviews the representative work in unsupervised HSI-CD and introduces the related background knowledge.

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In Section III, the details of the proposed method are introduced, including the architecture of the network, graph augmentation strategies, and design of UA contrastive loss. To evaluate the proposed method, the experiments on four HSI datasets are carried out and the related comparison with the SOTA methods as well as the discussion of results are illustrated in Section IV. Finally, Section V draws the conclusion of our work in this article.

II. Related Work

A. Hyperspectral Image Change Detection

For HSI-CD, the key problem is how to take the advantages of the rich spectral features and the context information to separate the changes from the unchanges [1]. In the early HSI-CD research, pixel-level statistical methods were mainly used to extract both spectral attributes and spatial context information contained in independent pixels and their neighbors. CVA, which determines the final changes through threshold calculation based on similarity measurement, is a common CD method. Unfortunately, CVA-based approaches have exposed certain drawbacks when dealing with HSI-CD due to the large number of bands for HSIs. To relieve the problem of data redundancy, PCA, which tries to map the original features linearly into a specified dimension, was widely used for HSI dimension reduction. Sparse PCA [25] and sparse representation [26] were applied to HSI-CD. Slow feature analysis [27] extracted the most temporally invariant components from the multitemporal images to transform the spectral data into a new feature space for HSI-CD. In addition, multiple morphological profiles [41] and three-order tucker decomposition [12] were also employed to effectively utilize the spectral–spatial information for HSI-CD. Although these methods based on traditional machine learning have made a certain contribution to HSI-CD, they also have obvious shortcomings, such as neglecting the spatial–spectral correlations among bands and the intrinsic characteristics of complex HSIs. Hasanlou and Seydi [6] analyzed the advantages and disadvantages of various classical methods in detail.

In recent years, with the rapid development of artificial intelligence techniques, deep learning has shown great potential in HSI-CD. In [28], fully convolutional network was employed to learn powerful spectral–spatial features for unsupervised HSI-CD. For full exploitation of the spectral information of HSIs, Wang et al. [14] proposed a general end-to-end 2-D CNN to detect changes from bi-temporal HSIs. Zhao et al. [15] designed a 3-D autoencoder to obtain spectral–spatial features and classified the changes from the unchanges via a 2-D CNN. Song et al. [29] combined 3-D fully convolutional network and long short-term memory to jointly extract spectral–spatial–temporal features for HSI-CD. In [30], hyperspectral change tensor was designed to make full use of the spectral–spatial information. Deep slow feature analysis [31] was proposed to suppress the unchanged components and highlight the changed components of the transformed features. More recently, attention-based methods have been developed for HSI-CD by taking advantage of the most informative spatial–spectral features from the huge-volume HSI data [32], [33], [34], [35]. In [33], a spectral–spatial wise attention was designed to extract discriminative spectral and spatial features but to suppress less informative ones. Along with attention mechanism, transformer had been applied for HSI-CD [36], [37], [38]. For instance, Wang et al. [36] developed a joint spectral, spatial, and temporal transformer for HSI-CD. However, most of these deep learning-based approaches depend heavily on a large set of training samples, which may be insufficient or even unavailable under certain circumstances. Considering this issue, pseudo labels were generated using the traditional CD approaches to train a CNN model to relieve the stress caused by limited available training HSI data [40]. To deal with the problem that only training samples are available in a single-time image, Tong et al. [42] employed transfer learning to classify the post-event image for CD. Though the drawback that deep learning-based methods mostly driven by annotated data suffer from expensive sample collection has been taken seriously recently [43], how to make full use of spatial and spectral information from HSI data to enhance the discriminative power of learned features is still a research priority currently [44], [45].

Most recently, to deal with the problem of the limited training samples while taking advantage of both spectral and spatial features, SSL-based HSI-CD methods have been reported [37], [46], [52]. In [39], Hu et al. proposed a self-supervised hyperspectral spatial–spectral feature understanding network by designing a spatial–spectral attention module to fully exploit the informative spatial and spectral features. Leenstra et al. [47] built a pretrained discriminative model to extract features from bi-temporal images for CD by making full use of pretext tasks. Saha et al. [48] integrated CL and deep clustering for multitemporal and multisensor CD, under the assumption that the changes are much less than the unchanges. Based on the CL [21], this method was capable of learning feature representation from input by pulling together the positive sample pairs and pushing away the negative sample pairs without any label. To generate positive and negative pair-wised samples, Cai et al. [49] exploited image segmentation to label the pixel pairs with the same category as positive samples and pixel pairs with different categories as negative ones. A similar method was proposed by Chen and Bruzzone [50]. They treated the pixel pairs located in the same superpixel as positive samples and used a teacher–student framework to estimate feature uncertainty to enhance the temporal robustness. Jiang et al. [51] proposed a global–local CL framework using image-level instance discrimination characteristics. To produce the pseudo training labels, Ou et al. [52] proposed a data augmentation strategy by adding random Gaussian noises to the difference image. In [53], an attention-based spatial and spectral network with PCA-guided self-supervised feature extraction was proposed for HSI-CD. However, these SSL-based HSI-CD methods still face the following limitations. First, the discriminative features of hyperspectral data should be preserved in SSL. To avoid the problem of the Hughes phenomenon, it is a common view that the high-dimensional hyperspectral data

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should be transformed into a low-dimensional space but the discriminative features, such as spectral attribute and topological structure, should be well preserved. Second, the uncertainty of learned features for downstream task should be concerned and delicately considered during the model design for SSL. In practice, structure uncertainty often occurs in graph construction where two nodes regarded as the nearest neighbors and connected by an edge may have different labels, which may further result in model uncertainty for downstream task. Third, data augmentation as the core of SSL should be treated seriously in consideration of correlation information between multitemporal HSIs.

B. Graph Embedding

Graph embedding [20], which is a popular way to find latent vector representation of graph but to preserve relevant network properties, acts as a bridge between structural and statistical approaches. Specifically, graph embedding tries to convert a graph into a low-dimensional space in which the graph structural information is preserved. By representing a graph as a low-dimensional vector, graph-related algorithms can then be computed efficiently. Hence, it allows the remote sensing image processing task to benefit from the computational efficiency of the SOTA statistical models and tools along with the convenience and representational power of classical symbolic representations. In [54], [55], and [56], graph embedding was applied for HSI feature extraction to improve the discriminating power of features for classification. However, embedding graph into low-dimensional space is not a trivial work. There are still a lot of issues needed to be further studied. First, the lack of sufficient labeled training data in HSI-CD is not sufficiently noticed. Second, the uncertainties of the learned low-dimensional embeddings are not taken into consideration in downstream task. Third, the computation burden of graph embedding is not received enough attention.

C. Graph Self-Supervised Learning

SSL, which was first introduced in the field of robotics, is further developed in the machine learning community for recovering the whole or parts of its original input [24], [57]. SSL can be viewed as a branch of unsupervised learning since there is no manual label involved for semantic feature learning and the learned features from SSL are used for subsequent tasks where the amount of the annotated data is limited. This provides a promising alternative for supervised deep learning algorithms where a model is trained on large-scale labeled datasets that are unavailable or prohibitively expensive to collect. Hence, SSL to some degree alleviates the sample annotation bottleneck, which is one of the main barriers to the practical deployment of deep learning. The mainstream SSL methods are commonly summarized into two general categories: generative and contrastive [21]. The former aims to train an encoder to encode input into an explicit vector and a decoder to reconstruct input from explicit vector, while the latter tries to train an encoder to encode input into an explicit vector to measure the similarity between them [21].

Due to the ability of learning useful informative representations from unlabeled data, SSL is naturally harnessed for providing additional supervision and achieves impressive improvements on various graph learning tasks. GSSL, which is designed to employ SSL techniques for graph data, usually involves context information or topological structure representation learning [21]. GSSL can go back to the early studies on unsupervised graph embedding [58], [59]. GSSL has paved the way for unsupervised learning of graph embeddings that are transferred to a variety of downstream tasks [21]. For example, in [60], GSSL is employed for molecular property prediction. Wu et al. [61] designed a GSSL-based learning paradigm for recommendation system. However, GSSL is still a freshman in the field of remote sensing image processing and how to apply GSSL to HSI-CD is a practical study to investigate. In addition, how to design a suitable data augmentation strategy under graph framework to fit the downstream CD task is also a problem to be solved.

III. Method

Let us consider a pair of HSIs, $\boldsymbol{I} = \{\boldsymbol{I}_1, \boldsymbol{I}_2\}$ ($\boldsymbol{I} \in R^{M\times N\times C}$), acquired over the same geographical area but at different times. It is assumed that the images have been registered, and the size of every image is $M \times N$ pixels. The HSIs contain C channels. Without any training data, our goal is to generate a special label set $\boldsymbol{Y} = \{y(m, n), 1 \le m \le M, 1 \le n \le N, y \in \{0, 1\}\}$, where 0 represents the unchanges and 1 denotes the changes occurred between the image pairs $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$. To achieve this purpose, we elaborate the details about the proposed UA-GSSL.

The overall fiowchart of the UA-GSSL is shown in Fig. 1. First, two graphs are constructed on the oversegmented images to represent the spectral and spatial correlations on input multitemporal HSIs. Second, based on these graphical structures, novel node- and edge-level data augmentations are designed to enrich the contrastive sample pairs. Then, the augmented graph pairs containing both positive and negative samples are input into a dual-branch SSL contrastive network by maximizing the mutual information between a pair of low-dimensional feature embeddings. Fourth, the extracted feature vectors are input to train the network via minimizing combination of the contrastive loss and the UA loss, which encourages the discriminative and reliable features to play the more dominant role in feature representation. Finally, a change map is produced using the learned features. The details of the UA-GSSL are as follows.

A. Spectral–Spatial Graph Construction

The first step is to construct a spectral–spatial graph to represent both the spectral and spatial contextual information of multitemporal HSIs. Since HSIs usually contain a large number of pixels, the computation burden will be too huge to meet the practical applications if every pixel in HSIs is considered as a graph node in spectral–spatial graph construction. To address this problem, in our method, rather than concentrating on individual pixels, a small region is

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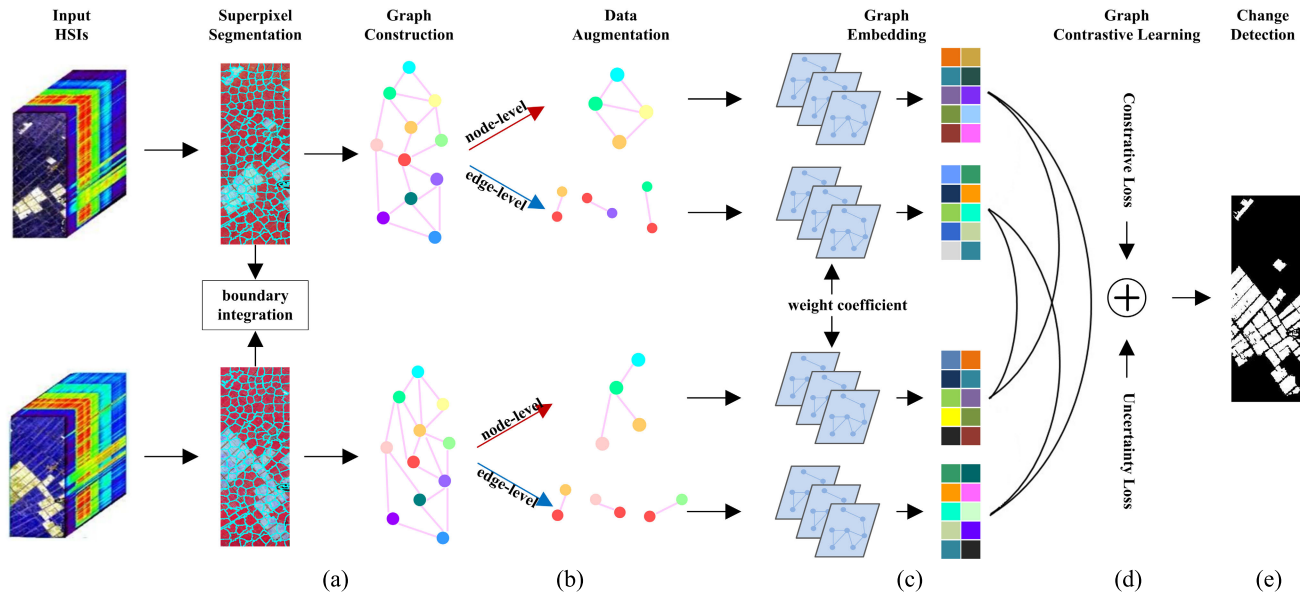


Fig. 1. Flowchart of the proposed UA-GSSL for HSI-CD: (a) superpixel segmentation and spectral–spatial graph construction, (b) node- and edge-level data augmentations, (c) graph embedding, (d) graph self-supervised contrastive network training, and (e) change map generation.

deemed as a basic unit in constructing a graph model on each temporal image.

Generally, image segmentation is a widely used way to generate spatially and semantically corresponding image regions for multispectral images. In the proposed method, a simple iterative cluster algorithm named SLIC [62] is employed and modified to generate a set of superpixels as the small regions for graph construction. Usually, a superpixel is composed of a group of pixels that are spectrally similar and spatially adjacent. Thus, these superpixels are highly homogeneous and compact [63]. Hence, all pixels lying in a superpixel can be viewed as a whole and they always have the same semantic label. In this article, every superpixel instead of pixel is considered as a node of the graph model and the relationship between the superpixels is viewed as the edge. The main reasons are given as follows. First, the use of superpixel greatly reduces the total number of basic units for multitemporal HSIs and thus significantly lowers the computational burden. Second, as superpixel is composed of a group of pixels in an image, it contains richer information including spectrum, texture, and spatial relationships with adjacent objects, which is particularly popular for remote sensing image interpretation. Third, superpixel is helpful for isolated noise suppression in remote sensing image CD. In this article, as the SLIC algorithm was not originally designed for HSIs, we modify the SLIC algorithm and enable it to be adaptive for multiple spectral channels of HSIs [64]. Based on this modified SLIC algorithm, each HSI is segmented. Unfortunately, as the changes and noises may occur between the multitemporal images, the produced superpixels, which are obtained by the superpixel segmentation algorithm carried out on each input multitemporal HSI separately, are not one-to-one at the same location on different temporal images. Thus, it is not feasible to directly compare the superpixels generated from the two input multitemporal images. In this article, we use the boundary integration strategy proposed in [63] to generate boundary-consistent superpixel segmentation maps for input

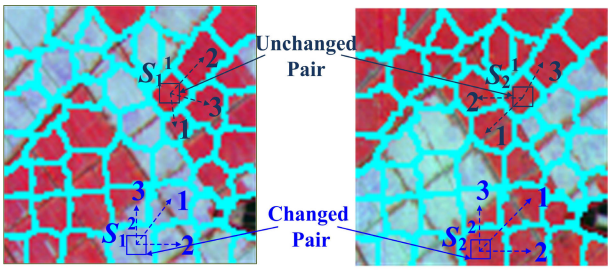


Fig. 2. Illustration of the graph models constructed on the superpixel pairs in multitemporal HSIs. The superpixel pairs may link with the completely different neighbors. Numbers 1, 2, and 3 indicate the three nearest neighbors.

HSI pair. In this way, for every superpixel s_1^i $(i = 1, 2, \ldots, n)$ (n is the total number of superpixels) in image $\boldsymbol{I}_1$, there is a corresponding superpixel s_2^i in image $\boldsymbol{I}_2$ at the same location.

Based on the two boundary-consistent superpixel segmentation maps, we construct two graph models, $\boldsymbol{G}_1 = (V_1, w_1)$ and $\boldsymbol{G}_2 = (V_2, w_2)$, on input HSI pair $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$, respectively. In graph construction, every superpixel s_t^i $(t = 1, 2)$ $(i = 1, 2, \ldots, n)$ is regarded as a vertex $v_t^i \in V_t$ $(t = 1, 2)$ $(i = 1, 2, \ldots, n)$, and an edge e_t^i is linked with the vertex v_t^i and its neighbor v_t^j $(j \neq i)$. Since superpixel segmentation aims to divide the pixels with similar spectral–spatial features into homogeneous regions, it is reasonable to regard the features of each superpixel as the feature vector of the graph node in graph construction. Noteworthy, although the two graph models $\boldsymbol{G}_t$ $(t = 1, 2)$ have the same number of nodes and every node pair poses the same location in the multitemporal images, the node pair on the same location may link with the completely different neighbors. As illustrated in Fig. 2, for the superpixel pair s_1^1 and s_2^1, they keep unchanged between the bi-temporal images $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$, but their three nearest neighbors are different and the graph models constructed on them are also different. Conversely, though changes are occurring between superpixel pair s_1^2 and s_2^2, they have three nearest neighbors on the same locations. Hence, the two graph models $\boldsymbol{G}_1$ and $\boldsymbol{G}_2$

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may have the wholly inconsistent topological structure, which enables the constructed graph models to elegantly characterize the spectral–spatial features of input multitemporal HSIs.

Based on the graph models $\boldsymbol{G}_1$ and $\boldsymbol{G}_2$, for every superpixel s_t^i in $\boldsymbol{I}_1$ or $\boldsymbol{I}_2$, it can be characterized by a spectral feature in the form of a vertex v_1^i or v_2^i and can also be represented by a spatial feature in the form of an edge e_1^i or e_2^i in graph model. In this way, every superpixel can be formally expressed with both the attribute features of node itself and the connectivity of nearby nodes in original input HSIs.

B. Graph-Based Data Augmentation

With the graph models constructed in the last step, GSSL is exploited to extract discriminative feature representations for downstream HSI-CD task in consideration of the issue that most deep learning-based supervised HSI-CD methods rely heavily on a large number of labeled training samples, but the training samples are scarce. CL is one of the most compelling methods for SSL and its main idea is to extract better representation for HSIs by grouping similar samples closer and enlarging different samples far from each other. To improve the generalization and robustness of CL, graph augmentation is adopted to explore underlying semantic and structural information. Recent studies [24] show that the success of CL relies heavily on well-designed data augmentation strategies. However, because of the inherent non-Euclidean properties of graph data, it is difficult to directly apply traditional data augmentations designed for images to graphs [24].

In this article, considering the nature of graph model, two node-level and edge-level data augmentation strategies are designed to expand the contrastive sample pairs.

1) Node-Level Data Augmentation: The node-level data augmentation mainly focuses on node attributes of graph model. The main idea of node-level data augmentation is that nodes with similar attributes, such as spectral features, should share similar representations and be assigned with the same labels. The node-level data augmentation strategy is defined as follows. With the constructed graph models $\boldsymbol{G}_1$ and $\boldsymbol{G}_2$, for a vertex v_1^i ($i= 1, 2, \ldots, n$) in $\boldsymbol{G}_1$ and its k-nearest neighbors $v_1^j \in N_{v_1^i}$ ($j \neq i$), there are corresponding vertex v_2^i ($i = 1, 2, \ldots, n$) in $\boldsymbol{G}_2$ and its k-nearest neighbors $v_2^j \in N_{v_2^i}$ ($j \neq i$). Under the prior probability of little changes occurring between multitemporal images [48], the vertices v_1^i and v_2^i are regarded as two different views of the same scene and the two vertices are considered as a positive sample pair. In addition, as a vertex v_2^j ($j \neq i$) in $N_{v_2^i}$ is the nearest neighbors of the vertex v_2^i, it should share similar representations and be assigned with the same label as v_2^i. Thus, the vertices v_1^i and $v_2^j \in N_{v_2^i}$ ($j \neq i$) and the vertices $v_1^j \in N_{v_1^i}$ ($j \neq i$) and v_2^i, are also viewed as positive sample pairs. Contrarily, the vertices v_1^i and v_2^k ($k \neq i$ and $k \neq j$) and vertices v_2^i and v_1^k ($k \neq i$ and $k \neq j$) are seen as negative sample pairs. In other words, under the CL framework, node-level data augmentation tries to pull the distance of node-level feature vectors between positive samples close and push the distance of node-level feature vectors between negative samples apart. In this article, the positive samples and negative samples obtained by node-level data augmentation are defined as $\boldsymbol{N}_p^v$ and $\boldsymbol{N}_n^v$, respectively.

2) Edge-Level Data Augmentation: The edge-level data augmentation principally concentrates on adjacency correlations of graph data. The edge-level data augmentation builds itself on the assumption that the vertices connected by an edge in a graph model will enjoy similar structural perspectives together and should be given the same labels. According to the definition of graph model, given a vertex v_t^i ($t = 1, 2$) ($i = 1, 2, \ldots, n$), there is an edge e_t^i linking this vertex and its neighbor v_t^j ($j \neq i$). Hence, the edge pair e_1^i and e_2^i for v_1^i and v_2^i, respectively, can be seen as a positive sample pair, while other edge pairs, such as e_1^i and e_2^j ($j \neq i$), are regarded as negative sample pairs. The edge-level data augmentation intends to keep the consistency of vertices within the same edge among multitemporal images close for positive sample pairs but apart for negative ones. In this article, the positive samples and negative samples achieved by edge-level data augmentation are defined as $\boldsymbol{N}_p^e$ and $\boldsymbol{N}_n^e$, respectively.

In this article, the positive and negative sample sets obtained by both node- and edge-level data augmentation strategies can be denoted as: $\boldsymbol{N}_p = [\boldsymbol{N}_p^v\ \boldsymbol{N}_p^e]$ and $\boldsymbol{N}_n = [\boldsymbol{N}_n^v\ \boldsymbol{N}_n^e]$, respectively.

C. Graph Embedding

According to the last two steps, multitemporal HSIs are represented by a set of graph-based positive and negative sample pairs, $\boldsymbol{N}_p$ and $\boldsymbol{N}_n$. However, the direct difference comparison operations on them are computationally expensive. In this article, a Siamese graph embedding network is designed to map the structure features for graph representations into embedding vectors so that the difference between the multitemporal HSIs can be reformed as the similarity measuring between embeddings with both the attribute features of superpixel itself and the correlations among nearby nodes.

In this article, a Siamese deep graph embedding network with a contrastive loss function is designed to embed every node v_t^i ($t = 1, 2$) ($i = 1, 2, \ldots, n$) in graph models $\boldsymbol{G}_t$ ($t = 1, 2$) as a low-dimensional embedding vector f_t^i ($t = 1, 2$) ($i = 1, 2, \ldots, n$) while preserving the graph structural information so that the high computation and space cost that most graph-based methods suffer can be greatly reduced. A ranking-based deep graph embedding technique, named Graph2Gauss [65], is adopted to represent each node in the form of a low-dimensional Gaussian distribution vector. The main idea of Graph2Gauss is to exploit the k-hop neighborhoods of an anchor node in the original graph model via a personalized ranking strategy. Formally, given an anchor node v_t^i, it is intuitively assumed that the node v_t^j belongs to the k-hop neighborhood of node v_t^i to be closer to v_t^i than the node v_t^k in the $k + 1$-hop neighborhood in terms of their embeddings, i.e., $\Delta(f_t^i, f_t^j) < \Delta(f_t^i, f_t^k), \forall j, k \in N_{v_t^i}, j < k$. To cater for the task of HSI-CD, a Siamese graph embedding network is adopted. As illustrated in Fig. 1, the Siamese graph embedding network consists of two identical Graph2Gauss networks sharing the same weights. Given the graph model $\boldsymbol{G}_t$ ($t = 1, 2$), the feature set fn_t for every node in node set V_t and the adjacency matrix A_t are the inputs of Graph2Gauss networks.

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Thus, for every node v_t^i $(t = 1, 2)$ $(i = 1, 2, \ldots, n)$, there is an output embedding $f_t^i = \{u_t^i, \Sigma_t^i\}$ $(t = 1, 2)$ $(i = 1, 2, \ldots, n)$ obtained via Graph2Gauss, where u_t^i and Σ_t^i are the mean and variance of Gaussian distribution, respectively.

D. Graph Contrastive Learning

According to the CL technique, when the positive dataset and the negative dataset, $\boldsymbol{N}_p$ and $\boldsymbol{N}_n$, are constructed, the CL pulls the positive pairs in $\boldsymbol{N}_p$ to be compact while pushing the negative pairs in $\boldsymbol{N}_n$ to be separable. Inspired by this idea, a graph-structured contrastive (GSC) loss function and a UA loss are novelly designed to guide the discriminative and reliable features to play a dominant role in downstream CD task.

1) Graph-Structured Contrastive Loss: A common way for CL is performing the contrastive process between the representations of two augmented views, i.e., enlarging the agreement between representations of the positive pairs while weakening that of the negative pairs. However, most of the previous studies contrasted the two views but neglected the structures of graph model, which captures limited characteristics. Thus, we argue that contrasting two graphs at both node level and edge level has the potential of capturing more abundant characteristics of graphs. To this end, we design a GSC module and define a loss function as follows:

$$L_1 = \frac{\sum_{j\in N_p^v} \mathrm{dis}\left(v_1^i, v_2^j\right)}{\sum_{k\in N_n^v} \mathrm{dis}\left(v_1^i, v_2^k\right)} + \beta_1 \frac{\sum_{j\in N_p^e} \mathrm{dis}\left(e_1^i, e_2^j\right)}{\sum_{k\in N_n^e} \mathrm{dis}\left(e_1^i, e_2^k\right)} \tag{1}$$

$$\mathrm{dis}\left(e_1^i, e_2^j\right) = \sum_{v_t^p \in e_t^i, v_t^q \in e_t^o} \mathrm{dis}\left(v_t^p, v_t^q\right), \quad (o = j, k), \ (t = 1, 2) \tag{2}$$

$$\mathrm{dis}\left(v_1^i, v_2^j\right) = \left(\mu_1^i - \mu_2^j\right)^T \left(\left(\sum_1^i\right)^{-1} - \left(\sum_1^j\right)^{-1}\right)\left(\mu_1^i - \mu_2^j\right) \tag{3}$$

where u_t^i and u_t^j, and Σ_t^i and Σ_t^j $(t = 1, 2)$ $(j \neq i)$ are the mean and variance of a pair of nodes v_t^i and v_t^j $(j \neq i)$ in graph models $\boldsymbol{G}_t$ $(t = 1, 2)$, respectively. β_1 in (1) is a weight coefficient to balance the two parts. The graph-structured contrastive loss aims to promote intraclass compactness and interclass separability by exploiting the prediction outcomes of unlabeled nodes, which supports the discriminative embeddings to act a leading role in change feature representation.

2) Uncertainty-Aware Loss: The contrastive loss function defined in the last step, however, does not consider the uncertainty caused by inexact graph structure during the spectral–spatial graph construction. For example, because of the phenomenon of "the same object with the different spectrum" and "the different object with the same spectrum," it is not sufficient enough to ensure that the k-nearest neighbors defined in graph construction will be assigned with the same label as that of its anchor, which inevitably causes the uncertainty representation in embedding learning for downstream HSI-CD task. For example, as shown in Fig. 2, the superpixel s_1^i and its third nearest neighbor may be assigned to different classes, which in turn would result in misleading neighbors and predictions. In addition, as the mixed pixels commonly exist in HSIs, the uncertainty problem is inherent for HSI-CD. Hence, it is desirable to incorporate the uncertainty of embedding representation into loss function design.

It is a common assumption in the remote sensing field that two objects are alike if they relate to similar objects. Hence, the nodes are similar if they pose the same role in the graph model. In other words, the superpixel pairs s_1^i and s_2^i in bi-temporal images $\boldsymbol{I}_1$ and $\boldsymbol{I}_2$, respectively, will keep similar or unchanged if their corresponding nodes v_1^i and v_2^i and their k-nearest neighbors v_1^j and v_2^j $(j \neq i)$ are all similar. Thus, the embeddings learned from the graph whose edges link the nodes with the labels same as that of its anchor will be compact and have a low variance. Contrarily, the learned embedding may have a large variance when the structure of constructed graph model is inexact. Therefore, the uncertainty of the learned feature representations for these nodes can be captured by the variances of these learned features, which is in line with the intuition that nodes in graph with less diverse neighborhoods have lower variance and usually belong to the same class, while nodes with high variance commonly have the different labels from their neighbors. Fortunately, as the embeddings learned in the last step inherently represent nodes in graph as Gaussian distribution, the variance can be exploited to represent the uncertainty of learned embeddings for downstream HSI-CD task. For example, when a node is located in a homogeneous region where most of the nodes located are assigned with the same label, the graph model constructed on it will be stable and the learned embeddings are expected to be robust. Hence, an UA module is designed in our method and the loss can be defined as

$$L_2 = \left(\mathrm{KL}\left(v_1^i, v_1^j\right) - \mathrm{KL}\left(v_2^i, v_2^j\right)\right) + \beta_2\left(\mathrm{KL}(e_1^i, e_1^j) - \mathrm{KL}\left(e_2^i, e_2^j\right)\right) \tag{4}$$

$$\mathrm{KL}\left(e_t^i, e_t^j\right) = \sum_{v_t^i \in e_t^i, v_t^j \in e_t^j} \mathrm{KL}\left(v_t^i, v_t^j\right), \quad (t = 1, 2) \tag{5}$$

$$\mathrm{KL}\left(v_1^i, v_1^j\right) = \frac{1}{2}\left[\mathrm{tr}\left(\Sigma_i^{-1}\Sigma_j\right) + \left(\mu_i - \mu_j\right)^T \Sigma_i^{-1}\left(\mu_i - \mu_j\right) - L - \log\frac{\det(\Sigma_i)}{\det\left(\Sigma_j\right)}\right] \tag{6}$$

where KL is the Kullback–Leibler (KL) divergence, $KL(v_t^i, v_t^j)$ is the KL divergence between nodes v_t^i and v_t^j $(t = 1, 2)$, $\mathrm{tr}(\cdot)$ denotes the trace of a matrix, and β_2 is a weight coefficient. One important property of UA loss is that if at least one of the two diagonal matrices has a large eigenvalue, the constraint that expects μ_i and μ_j to be close will be relaxed. Thus, this formulation naturally and explicitly integrates the uncertainty of the learned embeddings into the optimization of the graph self-supervised CL network, which encourages the reliable embeddings to act a main role in change feature representation.

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E. Change Detection

According to the contrastive loss and the uncertainty loss defined above, the overall loss function can be obtained

$$L = L_1 + \alpha^* L_2 \tag{7}$$

where α is a weight coefficient. Hence, the proposed Siamese graph self-supervised CL network can be trained by optimizing the loss function. After the training of graph self-supervised CL network, the trained network enables itself to act as the feature extractor to generate discriminative change feature representations. Hence, the correlation between the embedding vectors of HSIs obtained by the graph self-supervised CL network is helpful and crucial for the downstream HSI-CD task.

To fit into the CD task and produce the final change map, existing unsupervised CD algorithms can be employed to capture the change information by classifying the obtained embedding vectors $[|u_1 - u_2|\ |\Sigma_1 - \Sigma_2|]$ produced in the last step into two classes. It is regarded that the gained embeddings f_t^i for every superpixel in the last step is powerful enough for the downstream CD task. The larger the Euclidean distances are, the more certain changes occurred between the superpixel pairs. In this article, to illustrate the effectiveness of our proposed method, a simple unsupervised threshold algorithm, such as OTSU [66], is utilized to classify the distances because of its simplicity and practicality.

As the superpixel obtained in graph construction in Section III-A is a homogeneous and compact region, the pixels in a superpixel region usually belong to the same category. Hence, in this article, the label of a superpixel is directly assigned to the pixels in this superpixel. Also, the final pixel-level change map can be obtained.

IV. Experiments and Results

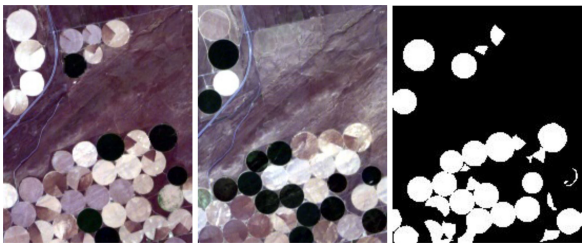
A. Study Area

Four HSI-CD datasets are used to evaluate the performance of the UA-GSSL. The details of the four datasets are as follows.

1) Hermiston Dataset: The Hermiston dataset[1] is composed of two HSIs acquired on May 1, 2004, and May 8, 2007, over Hermiston, OR, USA. The sensor used for shooting is Earth Observing-1 (EO-1) Hyperion, with a spectral range of 0.4–2.5 μm and a spatial resolution of 30 m. The size of the bi-temporal HSIs is 225 $\times$ 180 pixels, and 159 spectral bands with high SNR are selected for CD tasks. The covered land is a wide range of irrigated fields, river, and cultivated land. The pseudocolor images and ground-truth map are shown in Fig. 3. There are 9921 changed pixels and 30 579 unchanged pixels. The main change on this dataset is farmland land-cover change, including the transitions among crops, soil, water, and other land-cover types.

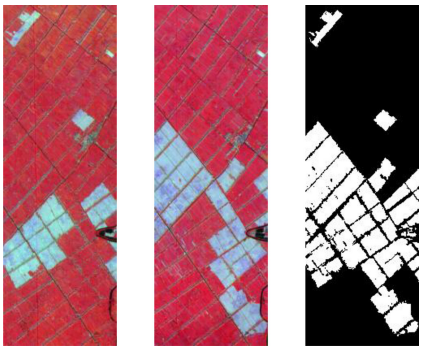
2) Farmland Dataset: The Farmland dataset[2] contains two HSIs taken on May 3, 2006 and April 23, 2007. The covered land is farmland in Yancheng, Jiangsu, China, as shown in Fig. 4(a) and (b). The sizes of the bi-temporal HSIs are both 420 $\times$ 140 pixels with a spatial resolution of 30 m acquired by the Earth Observing-1 (EO-1) Hyperion hyperspectral sensor. Both images in this dataset contain 155 bands covering the spectral range of 0.4–2.5 μm. After removing the noisy bands and water absorption bands, a total of 154 spectral bands are remained and used for CD. According to the ground-truth map shown in Fig. 4(c), the bi-temporal HSIs mainly include changes in corps and can be used in cultivated land change surveys. The numbers of changed and unchanged pixels are 18 277 and 40 523, respectively.

3) Barbara Dataset: The bi-temporal HSIs contained in the Santa Barbara dataset[3] were obtained in 2013 and 2014, with the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The covered land is Santa Barbara, CA, USA, as shown in Fig. 5(a) and (b). The main sceneries included in this dataset are mountainous and farmland. The two HSIs have the same size of 984 $\times$ 740 pixels with a spatial resolution of 20 m. Both HSIs in this dataset contain 224 spectral bands in the spectral



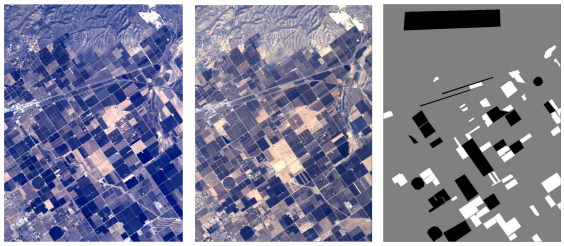
(a) (b) (c)

Fig. 3. Hermiston dataset. (a) Image acquired in 2004. (b) Image acquired in 2007. (c) Ground-truth image.



(a) (b) (c)

Fig. 4. Farmland dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground-truth image.



(a) (b) (c)

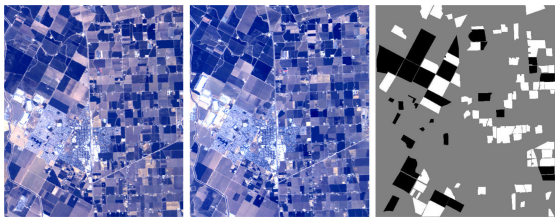
Fig. 5. Barbara dataset. (a) Image acquired in 2013. (b) Image acquired in 2014. (c) Ground-truth image.

[1]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[2]https://rslab.ut.ac.ir/data

[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

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(a) (b) (c)

Fig. 6. Bay dataset. (a) Image acquired in 2013. (b) Image acquired in 2015. (c) Ground-truth image.

range of 0.4–2.5 μm. The ground-truth image illustrated in Fig. 5(c) contains 52 134 changed pixels and 80 418 unchanged pixels, and the remaining pixels are unlabeled. The main changes for the two HSIs have recorded the dynamic changes of farmland.

4) Bay Dataset: The Bay dataset[4] contains a pair of HSIs taken in 2013 and 2015 by the AVIRIS sensor. It covers an area of Patterson, CA, USA, as shown in Fig. 6(a) and (b). The spatial dimensions are 600 × 500 pixels and both have 224 spectral bands in the 0.4–2.5-μm spectral region. This dataset is largely covered by farmlands and buildings. The ground-truth map shown in Fig. 6(c) is composed of 39 270 changed pixels, 34 211 unchanged pixels, and 226 519 undetermined pixels. The main changes for the two HSIs have recorded the urban evolution and dynamic changes of farmland.

For the Hermiston dataset and Farmland dataset, the white pixels in the ground-truth maps represent the changed regions, and the black pixels denote the unchanged regions. For the Bay dataset and Barbara dataset, the white pixels in the ground-truth maps denote the changed regions and the black pixels represent the unchanged regions, while the gray pixels are the unknown regions. In our experiments, we follow the conventions in previous methods and only the labeled changed and unchanged areas in the Bay and Barbara datasets are used for assessment.

B. Experiments Design

1) Comparison Methods: To testify the effectiveness of our proposed method, 12 excellent methods are selected for comparison, including CVA, GE-CVA, PCA, TDRD [6], DSFA [31], GETNET [14], CSANet [67], CSDBF [68], TriTF [69], Hypernet [37], MSCD [48], and S3Net [70].

1) **CVA** is an unsupervised HSI-CD method that obtains the change vectors and automatically classifies the changes and the unchanges via a K-means clustering algorithm [71].
2) **CVA** is an unsupervised HSI-CD method that obtains the change vectors and automatically classifies the changes and the unchanges via a K-means clustering algorithm [71].
3) **PCA** is an unsupervised HSI-CD method that produces the difference map for bi-temporal HSIs using the principal components obtained by a PCA algorithm.
4) **GE-CVA** is an unsupervised HSI-CD method that reduces the dimensionality of original hyperspectral data via a graph embedding algorithm [65] and discriminates the changed vectors from the unchanged ones using the CVA method. GE-CVA can be seen as a baseline of our proposed method.
5) **TDRD** is an unsupervised HSI-CD method using a three-order tucker decomposition to effectively exploit the spectral and spatial information contained in HSIs. The details can be found in this article [6] and code.[5]
6) **DSFA** is an unsupervised CD method that was deployed to suppress the unchanged components and highlight the changed components of the transformed features. The details can be found in this article [31] and code.[6]
7) **GETNET** is a supervised HSI-CD method founded on a general end-to-end 2-D CNN framework [14]. We implemented this approach using Python-3.7.
8) **CSANet** is a supervised HSI-CD method that extracts and integrates the joint spatial–spectral–temporal features by combining self-attention and CNN. The details can be found in this article [67] and code.[7]
9) **CSDBF** is a supervised CD method by combining graph attention network and CNN. In our experiment, we applied CSDBF for HSI-CD. The details can be found in this article [68] and code.[8]
10) **TriTF** is a supervised HSI-CD method based on a triplet transformer framework to fully mine the spatial–spectral information and temporal information for HSIs analysis. The details can be found in this article [69] and code.[9]
11) **HyperNet** is an SSL-based spatial–spectral feature representing algorithm for HSI-CD. The details can be found in this article [39] and code.[10]
12) **MSCD** is an SSL-based CD method originally designed for multisensor remote sensing images. MSCD trains a network in a self-supervised fashion by using deep clustering and CL. In our experiment, we applied MSCD for HSI-CD task. The details can be found in this article [48] and code.[11]
13) **S3Net** is an SSL-based CD method by exploiting both object-level spatial feature and the pixel-level spectral feature. The details can be found in this article [70] and code.[12]

The principle for choosing these algorithms is due to their relevance and popularity. Specifically, GE-CVA can be seen as a baseline of our proposed UA-GSSL. CVA, PCA, TDRD, DSFA, GETNET, CSANet, CSDBF, and TriTF are classical and popular CD algorithms and they employ different techniques to detect changes, such as image arithmetical operation, image transformation, and deep learning. Hypernet, MSCD, and S3Net are three recently

[4]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[5]https://github.com/zephyrhours/Hyperspectyral-Change-Detection-TDRD
[6]https://github.com/rulixiang/DSFANet
[7]https://github.com/SYFYN0317/CSANet
[8]https://github.com/zkylnnu/CSDBF
[9]https://github.com/zkylnnu/TriTF
[10]https://github.com/meiqihu/HyperNet
[11]https://gitlab.lrz.de/ai4eo/cd/tree/main/sarOpticalMultisensorTgrs2021
[12]https://github.com/OMEGA-RS/S3Net_CD

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reported self-supervised-based methods and can be regarded as related methods with the UA-GSSL.

2) Experimental Settings: We implemented our method UA-GSSL by Python-3.7 and using Tensorflow-2.1.0 as backend. The UA-GSSL is trained from scratch, and no pretrained model is used in our experiments. The network parameters are optimized using Adam with a fixed learning rate of 0.0001. The maximum number of epochs for training is set to 5000. The training and testing experiments were carried out on two NVIDIA RTX A6000 GPUs. Since region size is the main parameter of SLIC, in our UA-GSSL method, the parameters of region size are 3, 2, 5, and 5 for Hermiston dataset, Farmland dataset, Barbara dataset, and Bay dataset, respectively. Also, three nearest neighbors are adopted in all our experiments for graph construction. The weight coefficients α in formula (7), β_1 in formula (1), and β_2 in formula (4) are set as 0.5.

To make a fair comparison, in all the experiments, K-means clustering [71] is used as the automatic threshold selection for the unsupervised CD methods, including CVA, GE-CVA, and PCA. For the supervised deep learning-based methods, such as GETNET, CSANet, CSDBF, and TriTF, we adopt the same parameter settings introduced in the corresponding original papers and codes except for the number of selected training samples. In our experiments, 10% of the total data are randomly selected on the four datasets for training, and the remaining samples are used for testing. For the SSL-based methods, Hypernet, MSCD, S3Net, and UA-GSSL, no training example is needed and all the data are used as the testing samples.

In the prediction results of all methods, white represents changed areas and black represents unchanged areas.

3) Evaluation Criteria: To quantitatively evaluate the performance of different methods, seven most commonly used evaluation criteria are adopted in this article, including false alarm (FA) rate, miss alarm (MA) rate, overall accuracy (OA), Kappa coefficient (Kappa) [72], precision (Pre), recall rate (Recall), and F1 score (F1). All evaluation metrics used in the experiments are calculated as follows:

$$\mathrm{FA} = \frac{\mathrm{FN}}{\mathrm{TN}+\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{8}$$

$$\mathrm{MA} = \frac{\mathrm{FN}}{\mathrm{TN}+\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{9}$$

$$\mathrm{OA} = \frac{\mathrm{TN}+\mathrm{TP}}{\mathrm{TN}+\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{10}$$

$$P = \frac{(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})+(\mathrm{FP}+\mathrm{TP})(\mathrm{FN}+\mathrm{TP})}{(\mathrm{TN}+\mathrm{FN}+\mathrm{FP}+\mathrm{TP})^2} \tag{11}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-P}{1-P} \tag{12}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FP})} \tag{13}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FN})} \tag{14}$$

$$F1 = \frac{2*(\mathrm{Pre}*\mathrm{Recall})}{\mathrm{Pre}+\mathrm{Recall}}. \tag{15}$$

In their calculation, TP represents the true positive and TN denotes the true negative, indicating the number of changed and unchanged pixels that are correctly detected, respectively. FP is false positive, denoting that the unchanged pixels in the ground-truth map are erroneously detected as the changes in the change map, and FN is false negative, meaning that the changed pixels are misclassified as the unchanged class. FA and MA, which are two popular evaluation criteria for CD, indicate the false positive detection rate and false negative detection rate, respectively. OA index describes the proportion of predicted results consistent with the ground-truth labels, while Kappa [72] is employed to evaluate the consistency between the obtained HSI-CD results and the ground-truth maps. The higher the Kappa value is, the better the performance of the HSI-CD method achieves. F1, which is an indicator used in statistics to measure the accuracy of binary classification models, takes into account both Pre and Recall.

C. Experiment Results

1) Results on Hermiston Dataset: As shown in Fig. 7, the main change for Hermiston dataset is the size of farmland on this dataset. Compared with the reference change map in Fig. 3(c), the UA-GSSL is capable of detecting nearly all changes according to the visual observations. As the changed areas on Hermiston dataset are irregular and several changed parts are overlapped, this may lead to loss some useful edge information during superpixel segmentation. Hence, there are more miss or false detections near the boundary of changed areas compared with CVA, PCA, and TDRD methods. GE-CVA and PCA achieve a bit poorer performance than CVA because of dimension reduction, but a round unchanged area located on the top-left corner is falsely classified into the changes by CVA due to the noise. Compared with the change maps produced by other three self-supervised methods, Hypernet, MSCD, and S3Net, the UA-GSSL obtains a change map closer to the ground-truth map thanks to its UA loss design. With the training samples selected from the Hermiston dataset, the supervised methods, GETNET, CSANet, CSDBF, and TriTF, achieve better results than the other methods on the whole. Table I gives the quantitative evaluation of the different methods on the Hermiston dataset. After overall consideration of FA, MA, Kappa, F1, OA, Pre, and Recall, it can be obviously found that the UA-GSSL is an effective method for HSI-CD task. It acquires the best Kappa as 0.8464, F1 score as 0.8820, and OA as 0.9451 among all the compared self-supervised and unsupervised methods, which indicates that the UA-GSSL is comparable with these SOTA methods. With the respect to Kappa coefficient of the UA-GSSL, it means that the agreement between change map obtained by the UA-GSSL and the ground-truth map is almost perfect, which also demonstrates that the changes detected by the UA-GSSL are almost really changed areas. TDRD as an unsupervised approach gains the highest Pre with 0.9884 and PCA ranks second. The supervised method CSDBF, which is trained with 10% samples of total data, achieves the best results than all the other compared CD methods in terms of MA, OA,

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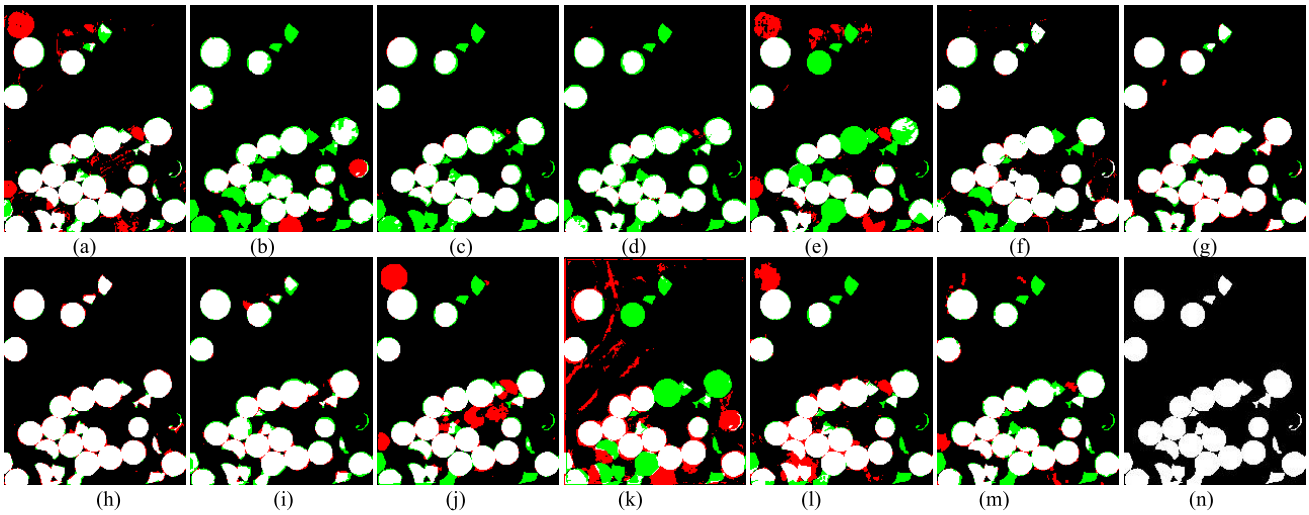


Fig. 7. CD results of different methods on the Hermiston dataset. The change maps are obtained by (a) CVA, (b) GE-CVA, (c) PCA, (d) TDRD, (e) DSFA, (f) GETNET, (g) CSANet, (h) CSDBF, (i) TriTF, (j) Hypernet, (k) MSCD, (l) S3Net, (m) UA-GSSL methods, and (n) reference map. The white areas in the images are the changes, while the black areas are the unchanges. FN and FP are labeled in green and red, respectively.

TABLE I

SUMMARY OF THE QUANTITATIVE EVALUATION FOR THE UA-GSSL AND OTHER SOTA METHODS ON THE HERMISTON DATASET. THE BEST RESULTS OF ALL METHODS ARE MARKED IN BOLD AND THE BEST RESULTS OF BOTH UNSUPERVISED AND SELF-SUPERVISED METHODS ARE MARKED WITH UNDERLINE

		FA	MA	Kappa	F1	OA	Pre	Recall
Un-Super	CVA	0.0438	0.0200	0.8331	0.8759	0.9362	0.8371	0.9185
	GE-CVA	0.0163	0.0786	0.7194	0.7781	0.9051	0.9109	0.6791
	PCA	0.0023	0.0635	0.8058	0.8466	0.9342	0.9875	0.7409
	TDRD	**0.0022**	0.0569	0.8272	0.8642	0.9409	**0.9884**	0.7677
	DFSA	0.0412	0.0961	0.5985	0.6844	0.8627	0.7832	0.6078
Super	GETNET	0.0075	0.0347	0.8817	0.9090	0.9579	0.9658	0.8585
	CSANet	0.0115	0.0137	0.9316	0.9483	0.9748	0.9527	0.9440
	CSDBF	0.0137	**0.0047**	**0.9509**	**0.9632**	**0.9816**	0.9461	**0.9808**
	TriTF	0.0101	0.0233	0.9081	0.9300	0.9666	0.9565	0.9048
Self-Super	Hypernet	0.0460	0.0346	0.7857	0.8394	0.9195	0.8207	0.8589
	MSCD	0.0882	0.0929	0.5072	0.6267	0.8189	0.6329	0.6207
	S3Net	0.0389	0.0307	0.8139	0.8602	0.9304	0.8462	0.8747
	UA-GSSL	0.0152	0.0397	0.8464	0.8820	0.9451	0.9309	0.8380

Recall, Kappa, and F1. However, DSFA does not do very well on this dataset. The reason may be that DSFA based on SFA theory is designed for linear data. On the whole, the UA-GSSL gets the best comprehensive performance on the Hermiston dataset according to both the qualitative and quantitative analyses. The proposed graph constrictive learning module and the UA learning module are targeted at utilizing both the spectral–spatial information contained in HSIs and the uncertainty into learned feature representation, beneficial to reduce the false detection and improve the CD, making a difference in the CD result on Hermiston dataset.

2) Results on Farmland Dataset: The size of farmland is the primary change on the Farmland dataset, and the changed information is related simple. Fig. 8 details the visual comparison of 11 different methods. It can be seen that the UA-GSSL correctly detects almost all the changes on the Farmland dataset. The main false detections of the UA-GSSL are many tiny and narrow unchanged areas located at the boundary of farmland, which are caused by superpixel segmentation in graph construction. As the boundary is the main uncertain area, the superpixel segmentation in turn deteriorates the uncertainty of graph model, which makes the discriminative and reliable features learning more difficult on the Farmland dataset. As a tensor-based method, the three-order tucker used in TDRD also renders the boundary of the achieved change map to be a little neat. Based on pixel-level features, CVA and PCA are capable of keeping the boundary of changed and unchanged areas more roughly than the UA-GSSL. Despite all these, the UA-GSSL gains a lower overall error rate than these traditional HSI-CD methods. For example, the change map illustrated in Fig. 8(l) indicates that the result produced by the UA-GSSL is more consistent with the ground truth than the CVA and PCA methods. Hence, it is not difficult to find that the UA-GSSL is able to detect more precise changes and capture most of the changes in an SSL way. As illustrated in Table II, the supervised methods GETNET, CSANet, CSDBF, and TriTF achieve the best change maps on the Farmland dataset, which is consistent with the results on the Hermiston dataset. Although these supervised deep learning-based methods reach more than 96% in terms of OA and Pre, the UA-GSSL, which is an SSL-based method, achieves comparable performance without any

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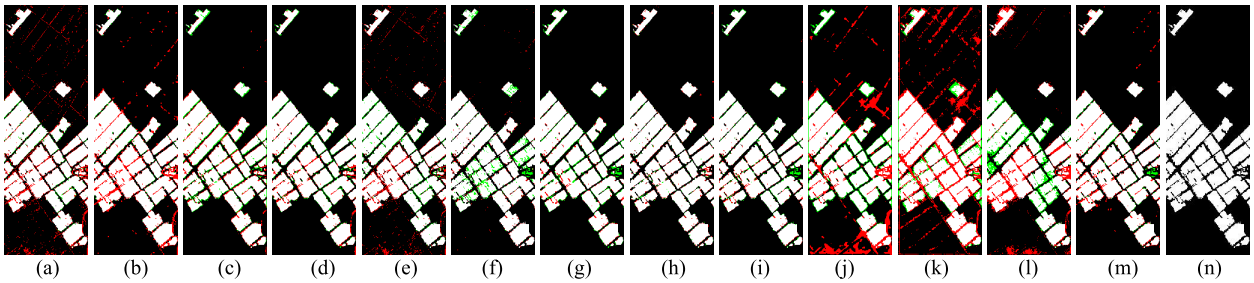


Fig. 8. CD results of different methods on the Farmland dataset. The change maps are obtained by (a) CVA, (b) GE-CVA, (c) PCA, (d) TDRD, (e) DSFA, (f) GETNET, (g) CSANet, (h) CSDBF, (i) TriTF, (j) Hypernet, (k) MSCD, (l) S3Net, (m) UA-GSSL methods, and (n) reference map. The white areas in the images are the changes, while the black areas are the unchanges. FN and FP are labeled in green and red, respectively.

TABLE II

SUMMARY OF THE QUANTITATIVE EVALUATION FOR THE UA-GSSL AND OTHER SOTA METHODS ON THE FARMLAND DATASET. THE BEST RESULTS OF ALL METHODS ARE MARKED IN BOLD AND THE BEST RESULTS OF BOTH UNSUPERVISED AND SELF-SUPERVISED METHODS ARE MARKED WITH UNDERLINE

		FA	MA	Kappa	F1	OA	Pre	Recall
Un-Super	CVA	0.0490	0.0072	0.8736	0.9154	0.9439	0.8611	0.9770
	GE-CVA	0.0467	0.0054	0.8827	0.9214	0.9479	0.8673	0.9827
	PCA	0.0253	0.0236	0.8858	0.9214	0.9510	0.9189	0.9239
	TDRD	0.0198	0.0148	0.9195	0.9447	0.9654	0.9372	0.9523
	DFSA	0.0346	0.0131	0.8906	0.9258	0.9523	0.8959	0.9577
Super	GETNET	**0.0094**	0.0216	0.9270	0.9492	0.9690	**0.9687**	0.9305
	CSANet	0.0115	0.0182	0.9303	0.9517	0.9703	0.9621	0.9415
	CSDBF	0.0123	**0.0051**	**0.9598**	**0.9724**	**0.9827**	0.9614	**0.9838**
	TriTF	0.0108	0.0136	0.9428	0.9605	0.9756	0.9648	0.9562
Self-Super	Hypernet	0.0747	0.0322	0.7596	0.8391	0.8931	0.7886	0.8964
	MSCD	0.1060	0.0161	0.7051	0.7880	0.8779	0.9338	0.6815
	S3Net	0.0508	0.0307	0.8132	0.8731	0.9186	0.8466	0.9013
	UA-GSSL	0.0245	0.0088	0.9233	0.9477	0.9667	0.9249	0.9718

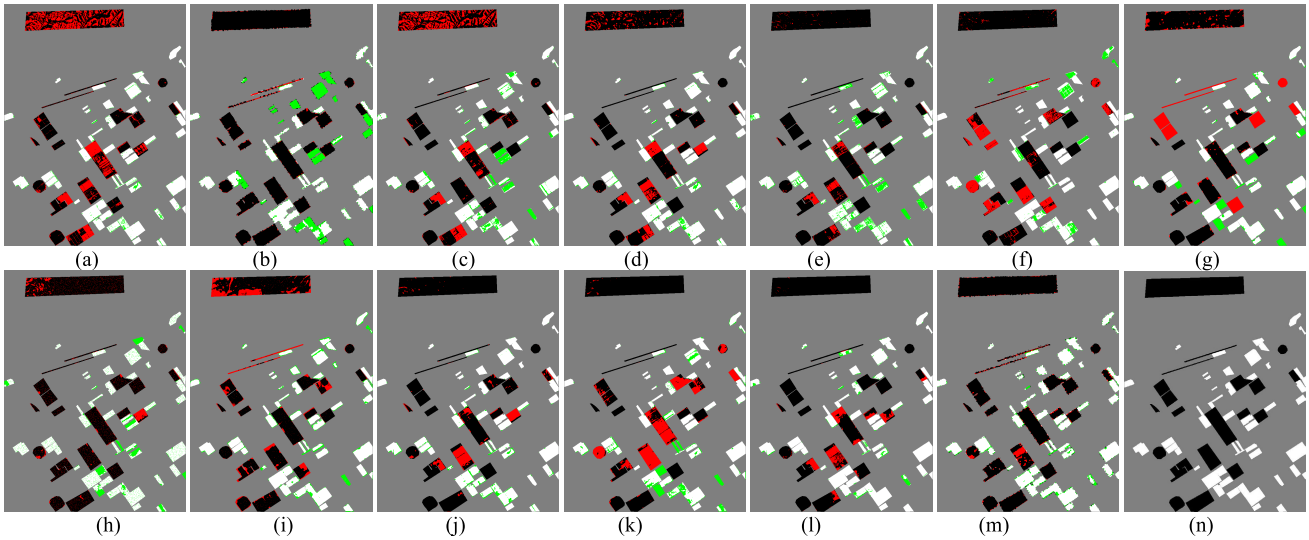


Fig. 9. CD results of different methods on the Barbara dataset. The change maps are obtained by (a) CVA, (b) GE-CVA, (c) PCA, (d) TDRD, (e) DSFA, (f) GETNET, (g) CSANet, (h) CSDBF, (i) TriTF, (j) Hypernet, (k) MSCD, (l) S3Net, (m) UA-GSSL methods, and (n) reference map. The white areas in the images are the changes, while the black areas are the unchanges. FN and FP are labeled in green and red, respectively.

training data. The UA-GSSL yields the highest Kappa, F1, OA, and Recall in comparison with all the self-supervised and unsupervised methods. Specifically, the UA-GSSL achieves the Kappa coefficient as 0.9233, F1 score as 0.9477, OA rate as 0.9667, and Recall rate as 0.9718, which are much higher than the other three SSL-based methods. This is primarily because of the input graph model, which allows encoding the spectral and spatial attributes in HSIs. In addition, the node- and edge-level data augmentations also enrich the contrastive sample pairs to take advantage of unlabeled data. To conclude, the UA-GSSL is capable of learning the discriminative and reliable features for HSI-CD thanks to its uncertainty awareness module.

3) Results on Barbara Dataset: On this dataset, the main changes for the two HSIs are the urban evolution and dynamic transformations of farmland. The CD results of different methods are shown in Fig. 9 and the quantitative evaluation on the Barbara dataset is listed in Table III. The uncertain area,

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TABLE III

SUMMARY OF THE QUANTITATIVE EVALUATION FOR THE UA-GSSL AND OTHER SOTA METHODS ON THE BARBARA DATASET. THE BEST RESULTS OF ALL METHODS ARE MARKED IN BOLD AND THE BEST RESULTS OF BOTH UNSUPERVISED AND SELF-SUPERVISED METHODS ARE MARKED WITH UNDERLINE

		FA	MA	Kappa	F1	OA	Pre	Recall
Un-Super	CVA	0.2356	**0.0263**	0.4983	0.7371	0.7382	0.6091	**0.9332**
	GE-CVA	**0.0309**	0.0604	0.5745	0.7089	0.8087	0.8829	0.5921
	PCA	0.1757	0.0833	0.4788	0.7053	0.7410	0.6383	0.7881
	TDRD	0.1181	0.0310	0.6993	0.8294	0.8509	0.7542	0.9211
	DFSA	0.0509	0.0961	0.6855	0.8017	0.8529	0.8537	0.7556
Super	GETNET	0.1621	0.0668	0.5399	0.7404	0.7711	0.6682	0.8301
	CSANet	0.1484	0.0745	0.5479	0.7410	0.7771	0.6824	0.8106
	CSDBF	0.0724	0.0795	0.6807	0.8052	0.8481	0.8125	0.7979
	TriTF	0.1187	0.0564	0.6430	0.7937	0.8249	0.7395	0.8566
Self-Super	Hypernet	0.0730	0.0327	0.7825	0.8722	0.8943	0.8317	0.9169
	MSCD	0.1699	0.0691	0.5114	0.7088	0.7611	0.8081	0.6313
	S3Net	0.0778	0.0279	0.7835	0.8737	0.8944	0.8245	0.9292
	UA-GSSL	0.0458	0.0458	**0.8080**	**0.8835**	**0.9084**	**0.8835**	0.8834

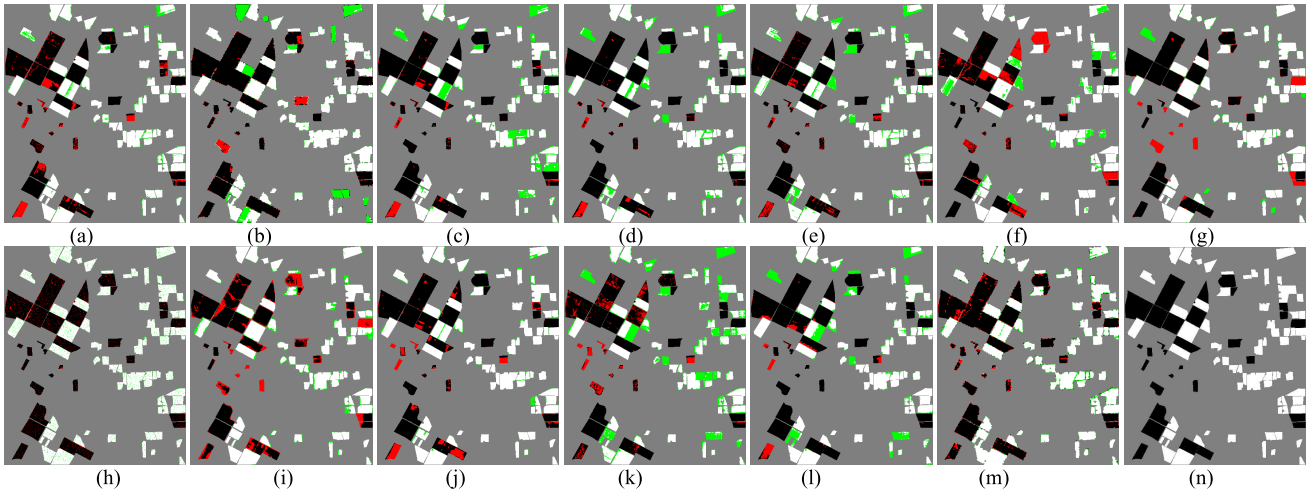


Fig. 10. CD results of different methods on the Bay dataset. The change maps are obtained by (a) CVA, (b) GE-CVA, (c) PCA, (d) TDRD, (e) DSFA, (f) GETNET, (g) CSANet, (h) CSDBF, (i) TriTF, (j) Hypernet, (k) MSCD, (l) S3Net, (m) UA-GSSL methods, and (n) reference map. The white areas in the images are the changes, while the black areas are the unchanges. FN and FP are labeled in green and red, respectively.

TABLE IV

SUMMARY OF THE QUANTITATIVE EVALUATION FOR THE UA-GSSL AND OTHER SOTA METHODS ON THE BAY DATASET. THE BEST RESULTS OF ALL METHODS ARE MARKED IN BOLD AND THE BEST RESULTS OF BOTH UNSUPERVISED AND SELF-SUPERVISED METHODS ARE MARKED WITH UNDERLINE

		FA	MA	Kappa	F1	OA	Pre	Recall
Un-Super	CVA	0.0734	0.0454	0.7603	0.8917	0.8812	0.8695	0.9150
	GE-CVA	0.0425	0.1112	0.6942	0.8464	0.8464	0.9088	0.7920
	PCA	0.0341	0.1266	0.6811	0.8354	0.8393	0.9229	0.7631
	TDRD	**0.0248**	0.0961	0.7595	0.8788	0.8791	**0.9464**	0.8203
	DFSA	0.0385	0.0955	0.7329	0.8676	0.8660	0.9194	0.8213
Super	GETNET	0.1085	0.0878	0.6042	0.8198	0.8036	0.8045	0.8356
	CSANet	0.0598	0.0422	0.7945	0.9061	0.8980	0.8916	0.9210
	CSDBF	0.0351	0.0378	**0.8536**	**0.9316**	**0.9271**	0.9339	0.9293
	TriTF	0.1166	0.0566	0.6491	0.8466	0.8268	0.8038	0.8942
Self-Super	Hypernet	0.0631	**0.0307**	0.8105	0.9148	0.9061	0.8886	**0.9425**
	MSCD	0.0665	0.1740	0.5214	0.7384	0.7595	0.6611	0.8362
	S3Net	0.0346	0.1141	0.7045	0.8497	0.8513	0.9240	0.7866
	UA-GSSL	0.0417	0.0467	0.8224	0.9169	0.9116	0.9213	0.9125

which is unlabeled, is in gray. As can be seen from the change maps in Fig. 9, it is obvious that the unsupervised approaches CVA, PCA, GE-CVA, and TDRD have plenty of false or miss detections appearing in the change maps, corresponding to low OA rate and Pre rate shown in Table III. The reason is that these methods were originally designed for linear data and are susceptible to the data source. The MSCD with a backbone of ResNet-34 is incapable of preprocessing the high-dimension data, which also causes it to perform not well on this dataset. The UA-GSSL is able to detect most of the changes under low

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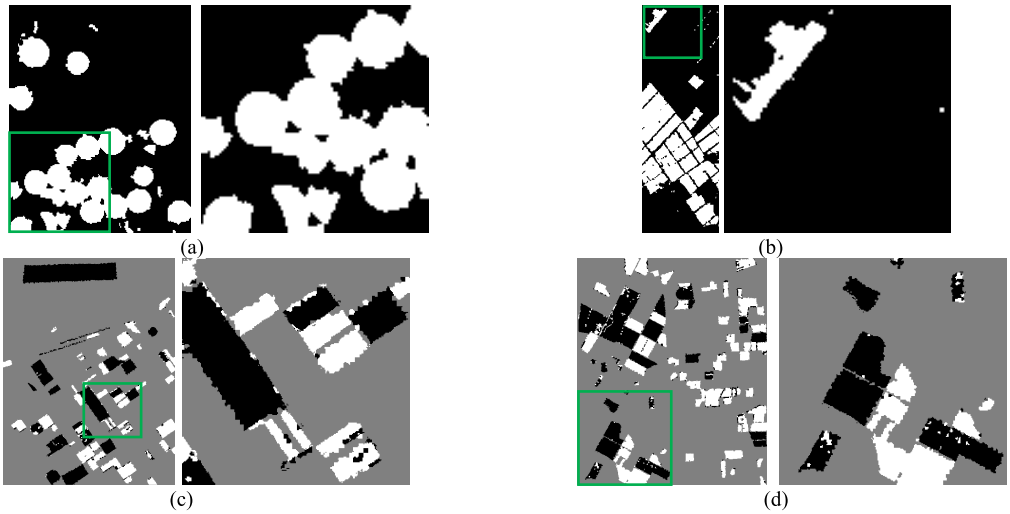


Fig. 11. Some details about CD maps obtained by our proposed method UA-GSSL on (a) Hermiston dataset, (b) Farmland dataset, (c) Barbara dataset, and (d) Bay dataset. The left images are change maps obtained by UA-GSSL and the right ones are enlarged details.

TABLE V

SUMMARY OF THE QUANTITATIVE EVALUATION ABOUT THE ABLATION EXPERIMENTS ON HERMISTON, FARMLAND, BARBARA, AND BAY DATASETS. THE BEST RESULTS OF ALL METHODS ARE MARKED IN BOLD AND THE SECOND BEST IS UNDERLINED

		FA	MA	Kappa	F1	OA	Pre	Recall
Hermiston	Base	0.0163	0.0786	0.7194	0.7781	0.9051	0.9109	0.6791
	Base + GSC	0.0216	0.0384	0.8339	0.8732	0.9400	0.9054	0.8432
	Base + UA	0.0537	**0.0322**	0.7745	0.8321	0.9141	0.7985	**0.8686**
	Base + GSC + UA	**0.0152**	0.0397	**0.8464**	**0.8820**	**0.9451**	**0.9309**	0.8380
Farmland	Base	0.0467	0.0054	0.8827	0.9214	0.9479	0.8673	**0.9827**
	Base + GSC	0.0290	0.0078	0.9156	0.9427	0.9632	0.9126	0.9749
	Base + UA	0.0457	**0.0047**	0.8865	0.9239	0.9496	0.8700	0.9850
	Base + GSC + UA	**0.0248**	0.0090	**0.9233**	**0.9471**	**0.9663**	**0.9242**	0.9711
Barbara	Base	**0.0309**	0.0604	0.5745	0.7089	0.8087	0.8829	0.5921
	Base + GSC	0.0449	0.0474	0.8060	0.8823	0.9077	**0.8852**	0.8794
	Base + UA	0.1162	0.0691	0.6198	0.7778	0.8147	0.7361	0.8244
	Base + GSC + UA	0.0458	**0.0458**	**0.8080**	**0.8835**	**0.9084**	0.8835	**0.8834**
Bay	Base	0.0425	0.1112	0.6942	0.8464	0.8464	0.9088	0.7920
	Base + GSC	0.0332	0.0594	0.8147	0.9112	0.9074	**0.9347**	0.8889
	Base + UA	**0.0287**	0.1240	0.6971	0.8432	0.8473	0.9346	0.7680
	Base + GSC + UA	0.0417	**0.0467**	**0.8224**	**0.9169**	**0.9116**	0.9213	**0.9125**

overall error. As presented in Table III, the UA-GSSL gains the best Kappa as 0.8080, F1 score as 0.8835, OA as 0. 9084, and Pre as 0.8835 among all comparative methods, while the CVA achieves the lowest MA as 0.0263 and the highest Recall with 0.9071. It is remarkable that the performance of supervised methods is not very satisfactory. As the scene on this dataset includes more ground objects than the above two datasets, the 10% training data may be not enough to train the deep networks. The UA-GSSL obtains the increase as high as 10%, 7%, 6%, 7%, and 2% in terms of Kappa, F1, OA, Pre, and Recall when compared with the best performance achieved by these supervised approaches, respectively. In conclusion, UA-GSSL is able to detect more precise CD results on the Barbara dataset, capturing subtle changes in an SSL way.

4) Results on Bay Dataset: On this dataset, there are some dynamic transformations on farmlands. Fig. 10 details the visual comparison of 12 different methods on the Bay dataset. The uncertain area, which is unlabeled, is in gray in the change maps in Fig. 10. It is observed that lots of false detections appear in the change maps of these four unsupervised methods, CVA, GE-CVA, PCA, and TDRD, corresponding to low OA rate shown in Table IV. Obvious missing and false detections can be found in the results. Remarkably, GETNET performs not well on this dataset. As GETNET heavily depends on the pseudo-training sets generated by CVA, however, these generated datasets may be insufficient to effectively train a deep network. The graph attention in CSDBF equips it well-suited to handle HSI-CD with limited samples, which empowers it to achieve the highest Kappa coefficient among all compared methods. S3Net, which is founded on superpixels, achieves a higher OA than UA-GSSL due to much less FAs. The UA-GSSL is able to detect most of the changes with low FA and MA. As illustrated in Table IV, the UA-GSSL gets the best Kappa as 0.8148, F1 as 0.9169, and OA as 0.9116 among all unsupervised and self-supervised methods, while the CSDBF gains the highest Kappa as 0.8536, F1 as 0.9316, and OA as 0.9271 among all comparative methods. Hence, it is not difficult to draw a conclusion that the UA-GSSL is effective to detect the changes on Bay dataset without any training samples.

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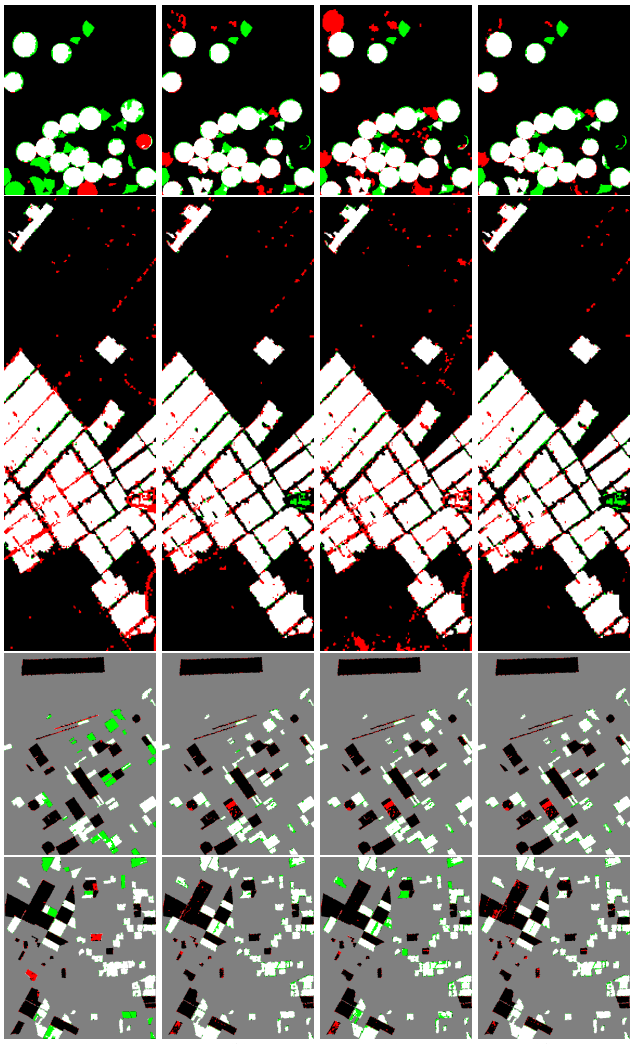


Fig. 12. CD results of ablation experiments. The change maps are obtained by (a) Base, (b) Base + GSC, (c) Base + UA, and (d) Base + GSC + UA methods on Hermiston, Farmland, Barbara, and Bay datasets. The white areas in the images are the changes, while the black areas are the unchanges. FN and FP are labeled in green and red, respectively.

In summary, the proposed method UA-GSSL achieves the satisfactory performance compared with all the other SOTA CD methods and is robust for different HSIs, which demonstrates the merits of the UA-GSSL. As shown in Fig. 11, some important details about our CD maps are illustrated for visual analysis. The virtues are mainly due to the novel graph-based node- and edge-level data augmentations and uncertainty-aware loss designs used in the UA-GSSL method.

D. Ablation Study

To test the performance of the proposed graph constrictive learning loss function and uncertainty-aware loss function on HSI-CD, extensive ablation experiments have been carried out to determine the roles of GSC module and UA module. Concretely, as the UA-GSSL method is founded on graph embedding algorithm, the "Base" in our experiment is chosen using GE-CVA. In the experiments, "Base + GSC" denotes that the GSC module is adopted compared with "Base," and "Base + UA" indicates that the UA module is employed

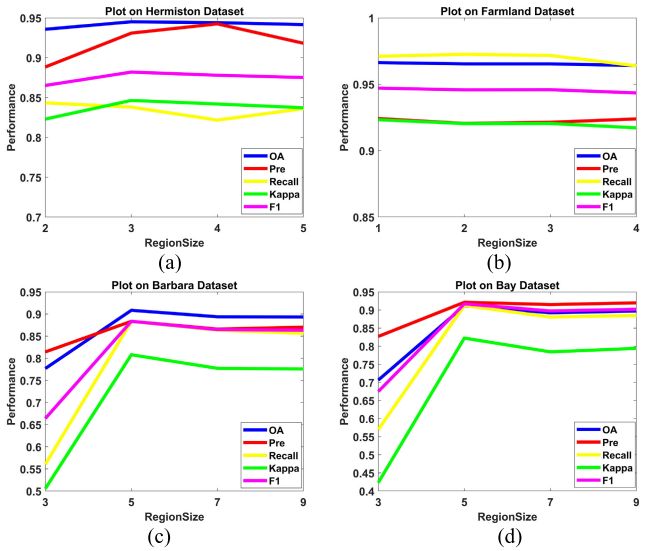


Fig. 13. Sensitivity analysis about the region size of superpixel segmentation versus the performance in terms of five quantitative evaluation indices OA, Pre, Recall, Kappa, and F1 on (a) Hermiston dataset, (b) Farmland dataset, (c) Barbara dataset, and (d) Bay dataset.

compared with "Base." "Base + GSC + UA" is then our proposed UA-GSSL that contains both GSC module and UA module.

Fig. 12 details the visual comparison of ablation experiments of the UA-GSSL and Table V gives the performance of ablation experiments on four datasets. The best results of all methods are marked in bold and the second best is underlined. "Base + GSC + UA" (the UA-GSSL) is almost always better than others across all four datasets. As shown in Table V, the "Base + GSC" and "Base + UA" gain better performance than "Base" does in most cases on four datasets. In particular, the improvement of "Base + GSC" is quite significant. For example, in comparison of the results of "Base," "Base + GSC" gains the increase at least 3%, 2%, 1%, and 4% in terms of Kappa, F1, OA, and Pre on four datasets, respectively. "Base + UA" also achieves distinct boost on Barbara and Bay datasets. "Base + GSC + UA," which is the UA-GSSL, acquires the best performance in terms of Kappa, F1, OA, Pre, and Recall on all four datasets.

Overall, according to the ablation experiments of the UA-GSSL, it is clear that the both GSC module and UA module play important roles in discriminative and reliable feature learning for downstream CD task. The GSC module, which heavily depends on the node- and edge-level data augmentations promotes the learned features to be distinguishable, while the UA module, which is designed to compensate for the uncertainty caused in graph construction encourages the reliable features to play a more dominant role in downstream CD task. All of these are quite helpful for discriminative feature learning, which greatly improves the performance of HSI-CD.

E. Stability Analysis of the Parameters

In this section, to validate the stability of the UA-GSSL, we analyze the effects of four parameters, weight coefficients

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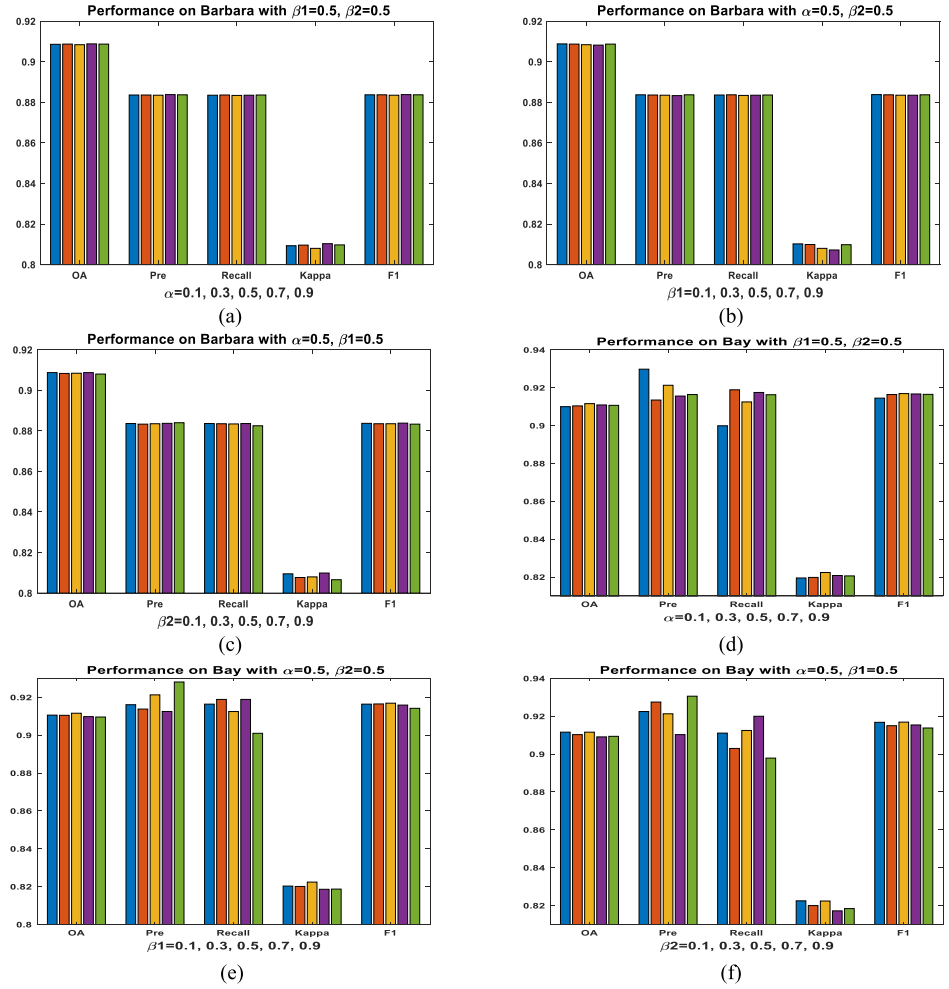


Fig. 14. Sensitivity analysis about the weight coefficients α, $\beta 1$, and $\beta 2$ versus the performance in terms of five quantitative evaluation indices OA, Pre, Recall, Kappa, and F1 on (a)–(c) Barbara and (d)–(f) Bay datasets.

TABLE VI

COMPARISON OF THE COMPUTATIONAL COST FOR SELF-SUPERVISED METHODS ON THE BARBARA DATASET

Method	Hypernet	MSCD	S3Net	UA-GSSL
Time (second)	478.35±6.33	280.31±0.28	1490.46±3.22	506.62±6.40

α in formula (7), β_1 in formula (1), β_2 in formula (4), and region size in superpixel segmentation, on performance of the UA-GSSL. Experiments are designed and conducted on four datasets and the results are evaluated with the terms of OA, Pre, Recall, Kappa, and F1. The details are as follows.

First, as the parameter of region size in superpixel segmentation is a key factor of the number of nodes N in graph construction, we aim to dig out the correlations between the parameter of region size and the CD performance. Concretely, based on the graph construction step of the proposed UA-GSSL, the parameter of region size is set as 2, 3, 4, and 5 for Hermiston dataset, as 1, 2, 3, and 4 for Farmland dataset, and as 3, 5, 7, and 9 for Barbara dataset and Bay dataset. Fig. 13 shows the relations about the region size of superpixel segmentation and the performance on four testing datasets. It is observed that the overall performance of CD improves first and then descends a little with the number of region size increasing on three datasets except the Farmland dataset. The best results are obtained on the Farmland dataset with a region size parameter setting as 1 and the curses in Fig. 13(b) drop a little with the region size climbing from 1 to 4. As the region size raises bigger, the number of nodes N in the graph model gradually gets smaller. However, the superpixels obtained by oversegmentation may contain some pixels with different labels when the region size becomes too big, which will cause the decline of CD performance. Fortunately, the curves eventually tend to be relatively stable. Hence, it is not difficult to draw the conclusion that the parameter of region size has a certain impact on the performance of the UA-GSSL, but this effect is not fatal.

Second, we design experiments about the three hyperparameters, α, β_1, and β_2, which are conceived mainly to exploit the influence on model's stability. Specifically, in our experiment, we set one of the hyperparameters α, β_1, and β_2 as 0.1, 0.3, 0.5, 0.7, and 0.9 and the other two hyperparameters as 0.5. We tested on four datasets. For limited resource and

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redundancies, we just report a part of experimental results in this article. As illustrated in Fig. 14, the performance on Barbara dataset keeps stable with the changes in parameters but a small fluctuation presents on the Bay dataset. The greatest fluctuation on the Bay dataset is about 0.02 in terms of Recall, while F1 and Kappa remain steady with different α, β_1, and β_2 values. Thus, it can be concluded that the performance does not fluctuate severely with the parameter changes, which indicates the robustness of the UA-GSSL. The main reason is that node-level augmentation and edge-level augmentation are to some extent complementary to each other.

Therefore, the experimental results and analyses validate that the UA-GSSL is stable and not sensitive to these hyperparameters.

F. Complexity Discussion

The time complexity for graph construction is $O(N^2)$, where N is the number of nodes in graph model. The time complexity of graph embedding is reported [65] as $O(K^2N)$, where K is the maximum distance considered and is usually a small value (for example, K is chosen as 3 in all our experiments). Thus, the time complexity of our proposed UA-GSSL is $O(N^2)$. Since the UA-GSSL constructs a graph model on superpixels instead of pixels, which largely reduces the number of nodes N, the training and testing time for the UA-GSSL is tolerable. Taking Barbara dataset (the largest multitemporal HSIs in our experiments) for example, the number of total superpixels and the nodes N in graph construction are 29 156 and 5246, respectively (the uncertain regions in Barbara dataset are not considered in graph construction, and thus, the two numbers are not equal). The time for graph construction (including superpixel segmentation), neural network training, and change labels prediction is 104.96 ± 3.13, 399.77 ± 16.01, and 1.89 ± 0.07 s. Hence, the total runtime of the UA-GSSL is about 506.62 s. The HyperNet takes 476.18 ± 12.65 and 2.17 ± 0.01 s for training and testing, respectively. The training time and testing time of the MSCD are 279.81 ± 0.55 and 0.50 ± 0.01 s, respectively. Table VI summarizes the computational cost for four self-supervised methods on the Barbara dataset. MSCD runs much faster than all other three methods on the Barbara dataset, while S3Net is most time-consuming. With a total of 5246 nodes, the UA-GSSL obtains the second largest time expense, and a main part of time consumption is spent on graph construction. Remarkably, the runtime cost of the UA-GSSL is highly dependent on the total number of superpixels, so reducing the number of nodes in graph is a way to reduce runtime. For example, as illustrated in the last section in Fig. 13(c), when the region size of superpixel segmentation is set as 7 on the Barbara dataset, the total number of graph nodes N decreases to 2672 and the total runtime of the UA-GSSL is about 56.00 s. The corresponding Kappa coefficient and F1 score are 0.7773 and 0.8648, respectively, which does not decrease seriously. Thus, the time cost of the UA-GSSL is acceptable in view of better comprehensive CD results.

V. Conclusion

In this article, we have put forward a graph self-supervised spatial–spectral feature learning network for HSI-CD. The proposed UA-GSSL employs an SSL mode and encodes not only the spectral and topological structure attributes but also the data uncertainty into learned feature representation for downstream CD task. The designed node- and edge-level data augmentations focus on the spectral and spatial correlation representation for multitemporal HSIs, while the uncertainty-aware loss concentrates on discriminative and reliable features learning in graph embedding. The performance of extensive experiments on four hyperspectral datasets demonstrates the effectiveness and generalization of the proposed UA-GSSL on HSI-CD task.

There are also some drawbacks for the UA-GSSL. First, the data augmentation used in the proposed UA-GSSL does not consider that different importance should be assigned to different nodes or edges. Second, the graph CL in the UA-GSSL only treats graph-level representation as a single contrastive objective and captures limited characteristics of graphs. Therefore, our further work will center on the above two lines.

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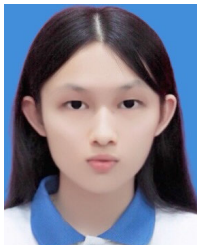
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